Gao 2021 Breast Cancer — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

After Harmony on sample, a single connected kNN graph and iLISI 1.00→1.14 (of max 5) read as 'integrated' BUT the embedding is organised BY PATIENT: per-patient self-neighbour enrichment 2.3–5.6× at θ=10; the worst GROWS with θ (6.0/5.6/6.0/7.6× over θ 5/10/20/40, so Harmony over-corrects rather than merges) and is PC-count invariant (5.6× at n_pcs 20/30/40). Every patient carries the same embryonic modules — identity, not biology. Clusters by sample: do not force a trajectory.

Gate 2 — module geography → NA

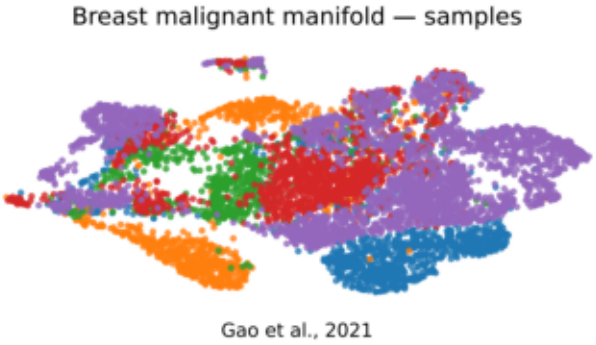
Not evaluated — Gate 1 NO-GO halts the chain before the trajectory / signature steps (a valid sample-integrated cell-state manifold is a precondition).

Gate 3 — hub cross-repression → NA

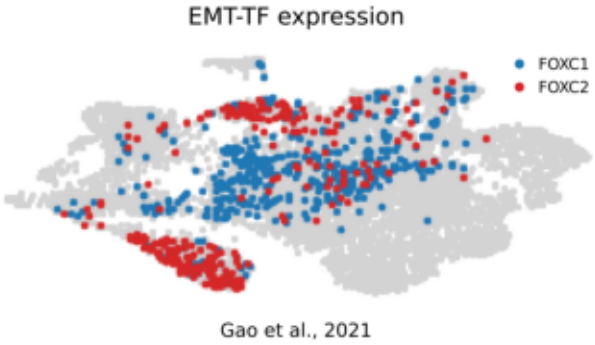
Not evaluated — branching-only, and no trajectory exists.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Triple-negative breast carcinoma + DCIS (Gao et al., 2021), 10x — 6,411 breast malignant cells (5 patients after threshold), scoped out of an 11-sample breast+thyroid bundle.

Verdict.
The pattern does NOT reproduce. The breast malignant compartment is a per-patient archipelago (inter-patient clonal / CNV heterogeneity), not a sample-integrated cell-state continuum. Gate 1 NO-GO — no trajectory fit; downstream notebooks not produced. Global module means Gast −0.024 / pSom +0.016 / Crest −0.042. Descriptive EMT-TF overlay: FOXC1 (28.5%) / FOXC2 (16.5%).

Parameter choices

Lineage scoping	scoped to breast on EPCAM/KRT8/18/19 high vs PAX8/TG~0 (bundle also had ATC thyroid)	**Sample threshold**	347 cells (gap 618→195, ratio 3.17); kept 5/6 breast samples → 6,411 cells (dropped TNBC3)
n_pcs / integration	n_pcs=20 (PC1=31%, inter-patient/CNV axis); Harmony θ swept 5/10/20/40; θ=10 best	**Geometry**	None — Gate 1 NO-GO, no manifold fit
Module means	Gast −0.024 / pSom +0.016 / Crest −0.042	**EMT-TFs**	FOXC1 (28.5%) + FOXC2 (16.5%); widened FOXC/PRRX; VIM excluded; descriptive